MEDSAFE AI

AI-Driven Medical Safety Assistant

Project Report

Department of Computer Science / Information Technology **[Vishwakarma Institute of Technology]** submitted in partial fulfillment of the requirements for the course **[Generative AI with Google Cloud]**

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Under the Guidance of
[Guide Name]

Academic Year: **2025–2026**

Certificate

This is to certify that the project titled **"MedSafe AI – AI-Driven Medical Safety Assistant"** submitted by **Pranav Ganorkar, Ankita Gaikwad, Aryaraj Wakhare, and Atharv Timane** is a bonafide work carried out under my supervision in partial fulfillment of the requirements for the course during the academic year **2025–2026**.

Project Guide	*Head of Department*	*Department*
Signature	*Signature*	*Seal*

Acknowledgement

We would like to express our sincere gratitude to our project guide for their continuous support, guidance, and encouragement throughout the development of this project.

We also thank the Department of Computer Science for providing the necessary infrastructure and resources to successfully complete the project.

Finally, we would like to thank our teammates for their cooperation and teamwork during the design and implementation of **MedSafe AI**.

Abstract

Healthcare systems face increasing challenges in ensuring medication safety, monitoring patient health conditions, and providing accessible medical guidance to individuals. Misinterpretation of prescriptions, lack of awareness of drug interactions, and delayed identification of abnormal health conditions can lead to serious health risks.

MedSafe AI is an intelligent healthcare assistance platform designed to improve medication safety awareness and provide early health risk insights through an integrated AI-driven system. The platform combines technologies such as machine learning, optical character recognition (OCR), real-time data streaming, and web-based dashboards to assist users in understanding medication interactions, analyzing symptoms, and monitoring physiological data.

The system allows users to upload prescription images, analyze potential drug interactions, log symptoms, and monitor simulated patient vitals in real time. Machine learning algorithms such as Isolation Forest and Autoencoder are used to detect anomalies in physiological data streams.

A real-time pipeline using Kafka processes incoming vitals data and stores results in a database for visualization and analysis. The system also includes a symptom guidance module and side-effect monitoring system to help users understand potential health conditions in an educational and non-diagnostic manner.

MedSafe AI aims to enhance medication literacy, improve early risk detection, and support preventive healthcare practices through an accessible AI-powered platform.

Table of Contents

Sr. No.	Title	Page No.
1	Introduction	01
2	Problem Statement	02
3	Objective	03
4	Literature Survey	04
5	System Architecture	05
6	System Design	06
7	Modules Description	07
8	Technologies Used	08
9	Implementation Methodology	09
10	Results & Output Screenshots	10
11	Advantages	15
12	Limitations	16
13	Future Scope	17
14	Conclusion	18
15	References	19

1. Introduction

Healthcare systems increasingly rely on digital technologies to improve patient safety and healthcare awareness. One of the most critical aspects of healthcare is **medication safety**, which involves ensuring that patients take medicines correctly and avoid harmful drug interactions.

Many patients struggle to understand prescriptions, recognize side effects, or determine whether symptoms require medical attention. These challenges often lead to incorrect medication usage, delayed treatment, and increased healthcare risks.

MedSafe AI is designed as an intelligent healthcare assistance system that helps users better understand medication usage and monitor potential health risks. The system integrates artificial intelligence, machine learning, and data streaming technologies to provide safety insights in a user-friendly interface.

By combining prescription analysis, drug interaction checking, symptom guidance, and anomaly detection in physiological data, MedSafe AI provides a comprehensive platform for healthcare awareness and monitoring.

2. Problem Statement

Medication errors and lack of health awareness are major issues in healthcare systems worldwide. Patients often face challenges such as:

- Difficulty understanding handwritten prescriptions
- Lack of awareness about harmful drug interactions
- Misinterpretation of symptoms
- Ignoring early warning signs of health conditions

These issues can lead to severe health complications if not addressed promptly.

There is a need for a system that can assist users in understanding medications, analyzing symptoms, and monitoring health conditions through intelligent algorithms.

3. Objectives

The primary objectives of MedSafe AI are:

- To develop an intelligent healthcare assistance platform
- To detect potential drug interactions between medicines
- To analyze prescriptions using OCR technology
- To monitor patient vitals using real-time data streams
- To detect abnormal health patterns using machine learning models
- To provide symptom-based guidance and health insights
- To generate alerts when abnormal conditions are detected

4. Literature Survey

Several healthcare monitoring systems have been developed to track patient health and manage medical records. However, many systems focus only on storing patient data rather than providing intelligent analysis.

Recent research has shown that machine learning algorithms can effectively detect anomalies in physiological data such as heart rate, blood pressure, and oxygen levels.

Algorithms such as **Isolation Forest** and **Autoencoders** are widely used for anomaly detection in healthcare datasets. These algorithms identify unusual patterns in data that may indicate potential health risks.

Optical Character Recognition technology has also been widely adopted in healthcare applications for extracting text from medical documents and prescriptions.

MedSafe AI integrates these technologies to create a unified healthcare assistance platform.

5. System Architecture

The architecture of MedSafe AI consists of multiple interconnected components that process data and generate healthcare insights.

Main components include:

Frontend		Dashboard
Flask	Backend	API
Kafka	Streaming	System
Machine	Learning	Models
Database		Storage
Alert Notification System		

Data Flow:

User Input → API Server → Data Processing → Machine Learning Analysis → Database → Dashboard Visualization

6. System Design

The system is designed as a modular architecture where each component performs a specific function.

Major components include:

- User Interface Layer
- Application Logic Layer
- Data Processing Layer
- Machine Learning Layer
- Data Storage Layer

This modular design improves scalability and maintainability of the system.

7. Modules Description

Dashboard Monitoring System

Displays patient vitals, anomalies, and alerts.

Prescription OCR Analyzer

Extracts medicine names and salts from uploaded prescription images.

Drug Interaction Checker

Detects harmful interactions between multiple medicines.

Symptom Guidance Engine

Provides basic health suggestions based on symptoms.

Side Effect Monitoring

Allows users to log post-medication experiences.

Real-Time Vitals Monitoring

Simulates patient vital signs data streams.

Anomaly Detection Module

Uses machine learning models to detect abnormal patterns.

8. Technologies Used

Technology	Purpose
Python	Core programming language
Flask	Backend API framework
React / Web UI	User interface
Apache Kafka	Real-time data streaming
Machine Learning	Anomaly detection
Tesseract OCR	Prescription text extraction
SQLite / PostgreSQL	Database storage

9. Implementation Methodology

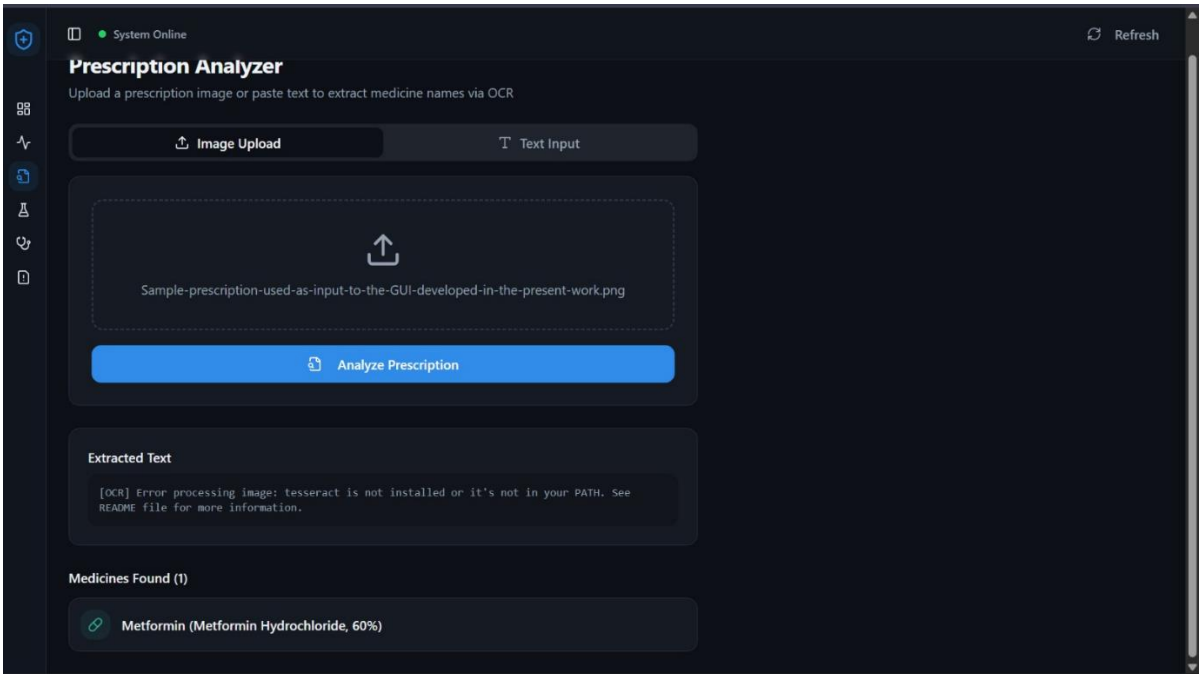
The implementation process involved several stages:

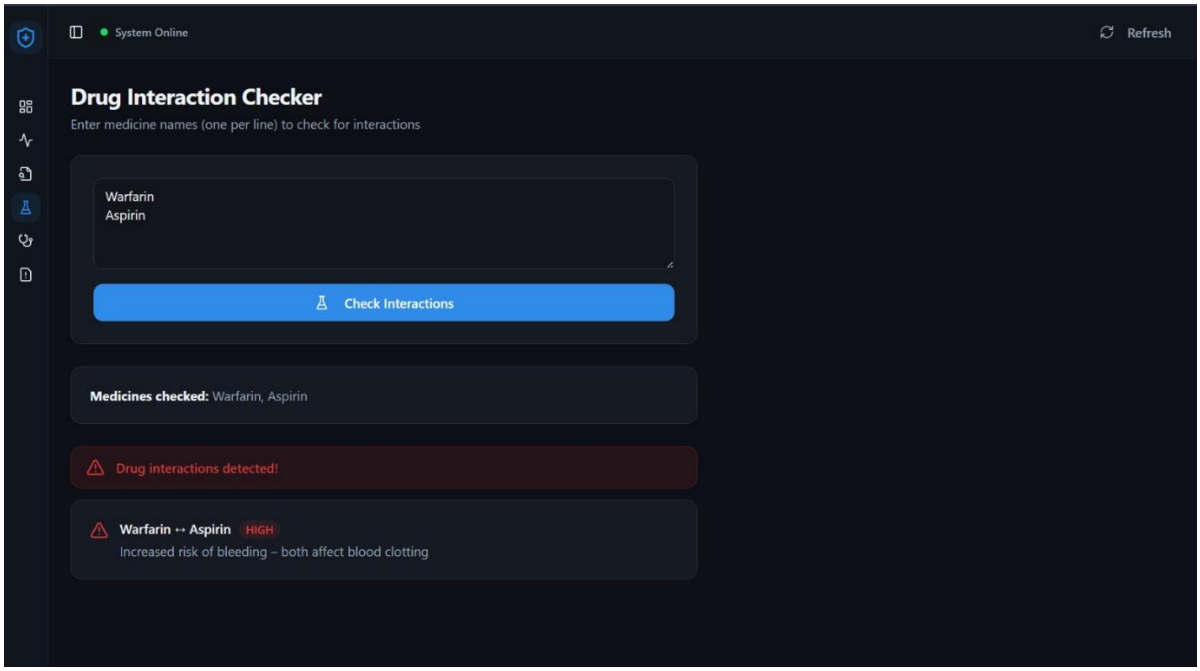
1. Designing the system architecture
2. Creating backend APIs using Flask
3. Implementing Kafka producer and consumer for real-time vitals data
4. Integrating machine learning models for anomaly detection
5. Developing the web-based user interface
6. Connecting database storage
7. Implementing alert and notification systems

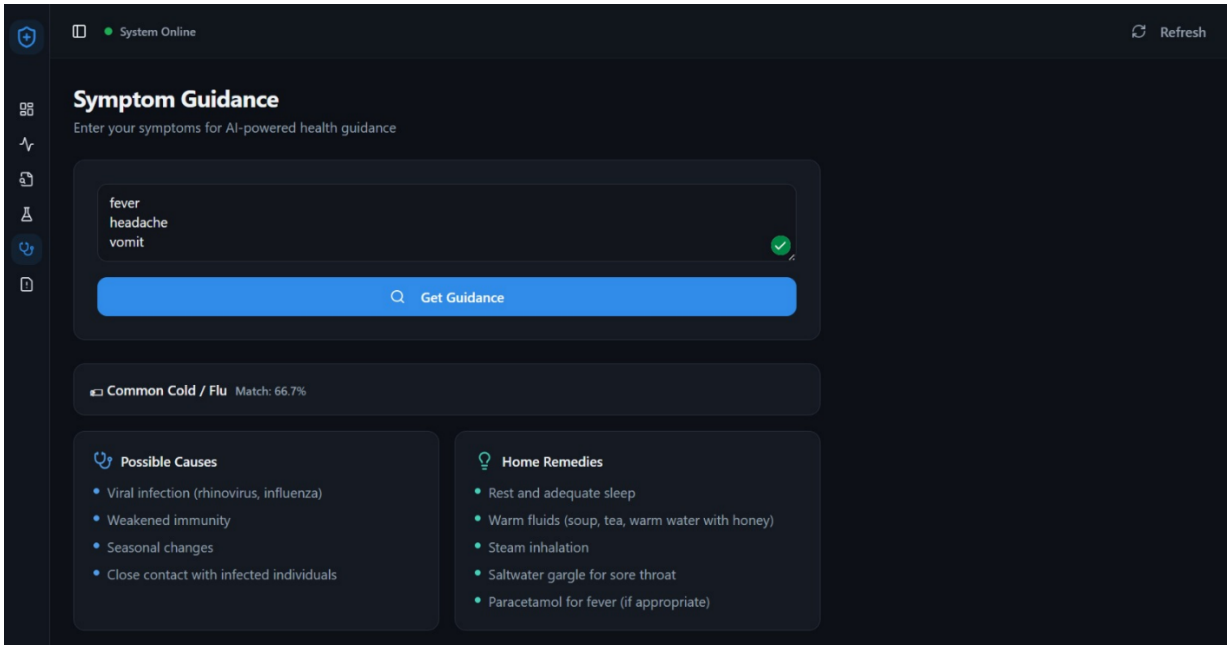
10. Results and Output

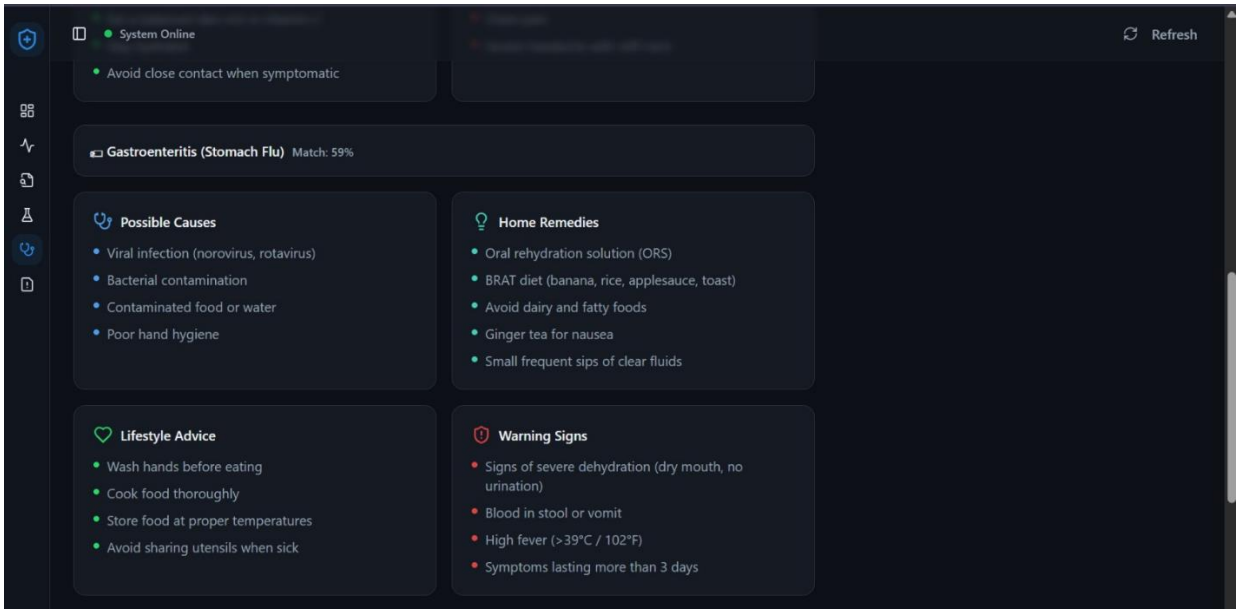
The system successfully demonstrates the following functionalities:

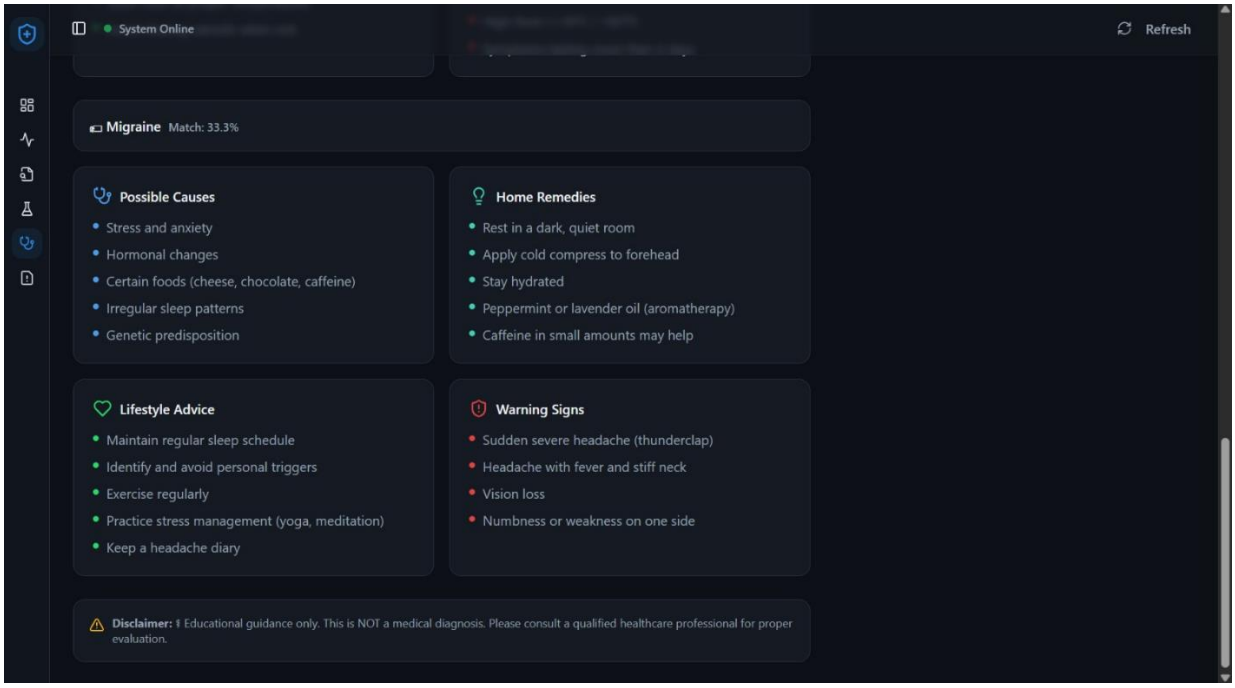
- Real-time vitals monitoring
- Drug interaction detection
- Prescription text extraction
- Symptom guidance suggestions
- Side-effect monitoring reports
- Machine learning based anomaly detection

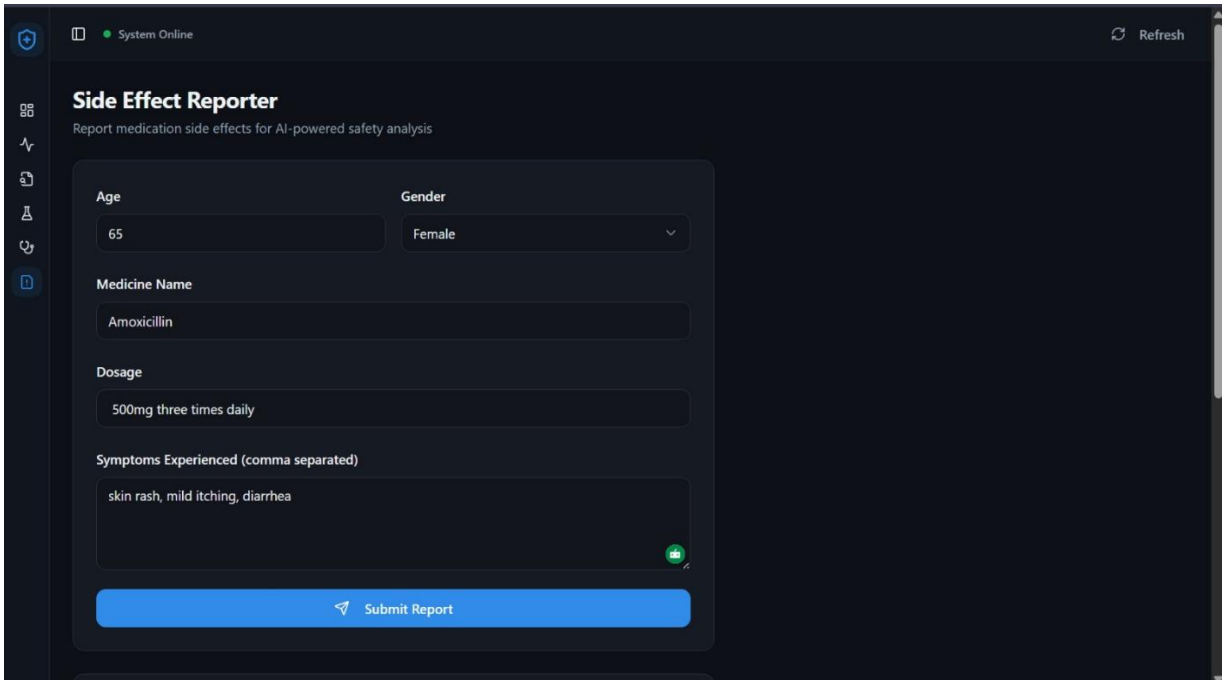

System Online
Refresh
Prescription Analyzer
Upload a prescription image or paste text to extract medicine names via OCR
Image Upload
Text Input
Sample-prescription-used-as-input-to-the-GUI-developed-in-the-present-work.png
Analyze Prescription
Extracted Text
[OCR] Error processing image: tesseract is not installed or it's not in your PATH. See README file for more information.
Medicines Found (1)
Metformin (Metformin Hydrochloride, 60%)

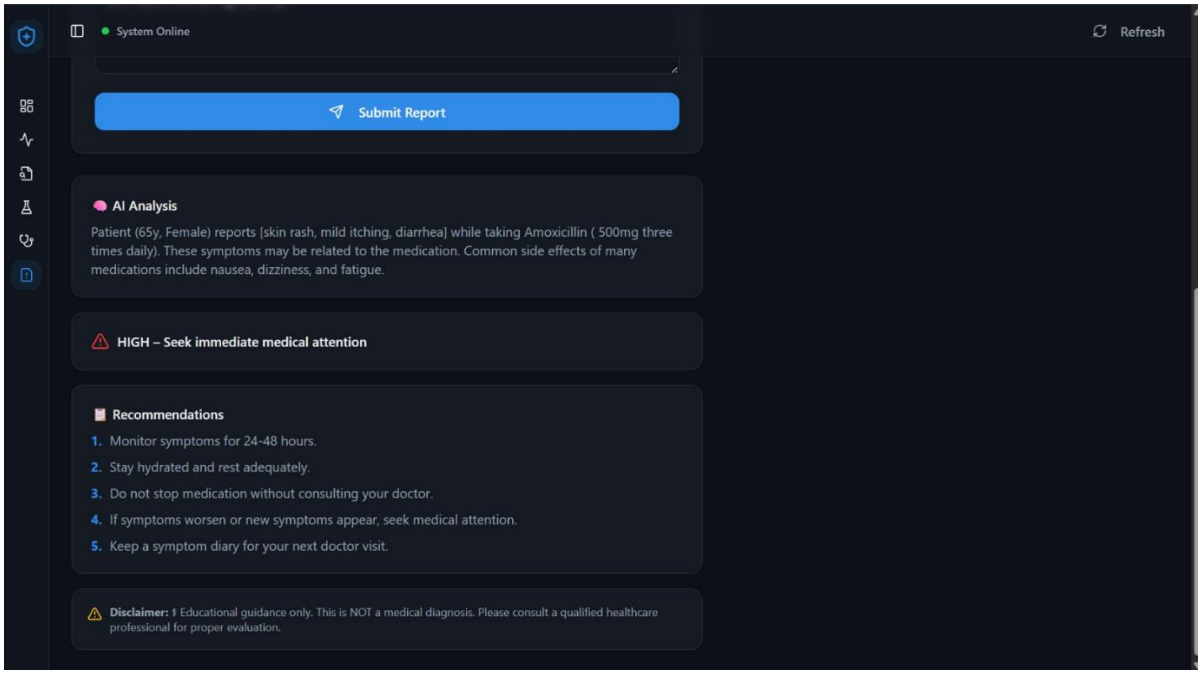

System Online
Refresh
Drug Interaction Checker
Enter medicine names (one per line) to check for interactions
Warfarin
Aspirin
Check Interactions
Medicines checked: Warfarin, Aspirin
Drug interactions detected!
Warfarin ↔ Aspirin HIGH
Increased risk of bleeding – both affect blood clotting


System Online
Refresh
Symptom Guidance
Enter your symptoms for AI-powered health guidance
fever
headache
vomit
Get Guidance
Common Cold / Flu Match: 66.7%
Possible Causes
Viral infection (rhinovirus, influenza)
Weakened immunity
Seasonal changes
Close contact with infected individuals
Home Remedies
Rest and adequate sleep
Warm fluids (soup, tea, warm water with honey)
Steam inhalation
Saltwater gargle for sore throat
Paracetamol for fever (if appropriate)


System Online
Refresh
Avoid close contact when symptomatic
Gastroenteritis (Stomach Flu) Match: 59%
Possible Causes
Viral infection (norovirus, rotavirus)
Bacterial contamination
Contaminated food or water
Poor hand hygiene
Home Remedies
Oral rehydration solution (ORS)
BRAT diet (banana, rice, applesauce, toast)
Avoid dairy and fatty foods
Ginger tea for nausea
Small frequent sips of clear fluids
Lifestyle Advice
Wash hands before eating
Cook food thoroughly
Store food at proper temperatures
Avoid sharing utensils when sick
Warning Signs
Signs of severe dehydration (dry mouth, no urination)
Blood in stool or vomit
High fever (>39°C / 102°F)
Symptoms lasting more than 3 days


System Online
Refresh
Migraine Match: 33.3%
Possible Causes
Stress and anxiety
Hormonal changes
Certain foods (cheese, chocolate, caffeine)
Irregular sleep patterns
Genetic predisposition
Home Remedies
Rest in a dark, quiet room
Apply cold compress to forehead
Stay hydrated
Peppermint or lavender oil (aromatherapy)
Caffeine in small amounts may help
Lifestyle Advice
Maintain regular sleep schedule
Identify and avoid personal triggers
Exercise regularly
Practice stress management (yoga, meditation)
Keep a headache diary
Warning Signs
Sudden severe headache (thunderclap)
Headache with fever and stiff neck
Vision loss
Numbness or weakness on one side
Disclaimer: Educational guidance only. This is NOT a medical diagnosis. Please consult a qualified healthcare professional for proper evaluation.


System Online
Refresh
Side Effect Reporter
Report medication side effects for AI-powered safety analysis
Age
65
Gender
Female
Medicine Name
Amoxicillin
Dosage
500mg three times daily
Symptoms Experienced (comma separated)
skin rash, mild itching, diarrhea
Submit Report


System Online
Refresh
Submit Report
AI Analysis
Patient (65y, Female) reports [skin rash, mild itching, diarrhea] while taking Amoxicillin (500mg three times daily). These symptoms may be related to the medication. Common side effects of many medications include nausea, dizziness, and fatigue.
HIGH – Seek immediate medical attention
Recommendations
1. Monitor symptoms for 24-48 hours.
2. Stay hydrated and rest adequately.
3. Do not stop medication without consulting your doctor.
4. If symptoms worsen or new symptoms appear, seek medical attention.
5. Keep a symptom diary for your next doctor visit.
Disclaimer: ⚕ Educational guidance only. This is NOT a medical diagnosis. Please consult a qualified healthcare professional for proper evaluation.

11. Advantages

- Improves medication safety awareness
- Provides early health risk detection
- Easy to use web interface
- Combines multiple healthcare tools in one platform
- AI-based anomaly detection

12. Limitations

- Uses simulated patient vitals data
- Limited medical dataset for drug interactions
- Not intended to replace professional medical advice

13. Future Scope

Possible future improvements include:

- Integration with wearable health devices
- Mobile application development
- Integration with hospital databases
- Advanced AI models for diagnosis
- Real patient health data integration

14. Conclusion

MedSafe AI demonstrates how artificial intelligence and modern software technologies can be used to enhance healthcare awareness and medication safety. By integrating real-time data streaming, machine learning algorithms, and intelligent healthcare modules, the system provides a comprehensive platform for monitoring health conditions and understanding medication risks.

The platform helps users interpret prescriptions, detect drug interactions, monitor symptoms, and identify abnormal physiological patterns. These features highlight the potential of AI-driven healthcare assistance systems in improving preventive healthcare and patient safety.

15. References

1. Machine Learning for Healthcare Monitoring – IEEE
2. Anomaly Detection Techniques in Healthcare Data
3. Apache Kafka Documentation
4. Flask Web Framework Documentation
5. Optical Character Recognition (OCR) Research Papers
6. Medical Drug Interaction Databases